```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
data=pd.read_csv('/content/Iris (1) - Iris (1).csv')
data
```


```
{"summary":"{\n  \"name\": \"data\",\n  \"rows\": 150,\n  \"fields\":
[\n    {\n      \"column\": \"sepal.length\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\":
0.8280661279778629,\n        \"min\": 4.3,\n        \"max\": 7.9,\n
\"num_unique_values\": 35,\n        \"samples\": [\n          6.2,\n
4.5,\n          5.6\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"sepal.width\",\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 0.435866284936698,\n        \"min\":
2.0,\n        \"max\": 4.4,\n        \"num_unique_values\": 23,\n
\"samples\": [\n          2.3,\n          4.0,\n          3.5\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    },\n    {\n      \"column\": \"petal.length\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
1.7652982332594667,\n        \"min\": 1.0,\n        \"max\": 6.9,\n
\"num_unique_values\": 43,\n        \"samples\": [\n          6.7,\n
3.8,\n          3.7\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"petal.width\",\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 0.7622376689603465,\n        \"min\":
0.1,\n        \"max\": 2.5,\n        \"num_unique_values\": 22,\n
\"samples\": [\n          0.2,\n          1.2,\n          1.3\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    },\n    {\n      \"column\": \"variety\",\n
\"properties\": {\n        \"dtype\": \"category\",\n
\"num_unique_values\": 3,\n        \"samples\": [\n
\"Setosa\",\n          \"Versicolor\",\n          \"Virginica\"\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    }\n  ]\n}","type":"dataframe","variable_name":"data"}
```


```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
```

```
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

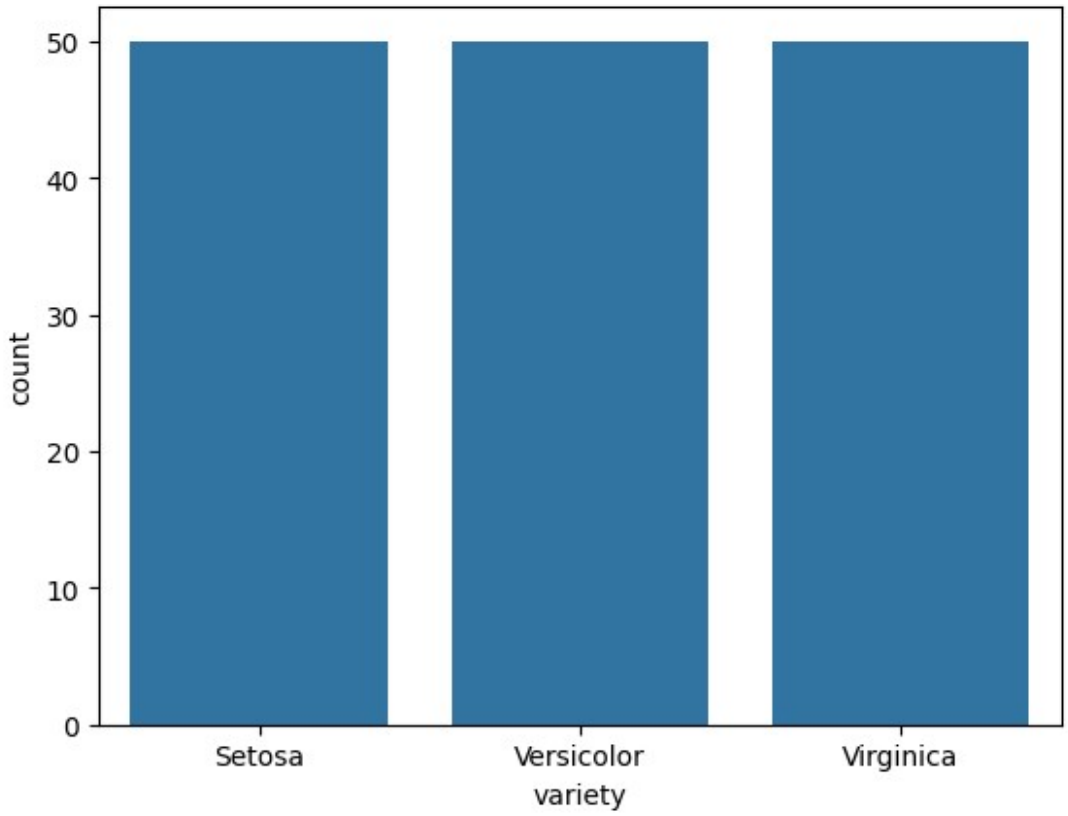
```
data.describe()
```


```
{"summary":"{\n  \"name\": \"data\",\n  \"rows\": 8,\n  \"fields\": [\
n    {\n      \"column\": \"sepal.length\",\n      \"properties\": {\n
\"dtype\": \"number\",\n        \"std\": 51.24711349471842,\n
\"min\": 0.8280661279778629,\n        \"max\": 150.0,\n
\"num_unique_values\": 8,\n        \"samples\": [\n
5.843333333333334,\n          5.8,\n          150.0\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"sepal.width\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
52.08617800869866,\n        \"min\": 0.435866284936698,\n
\"max\": 150.0,\n        \"num_unique_values\": 8,\n
\"samples\": [\n          3.057333333333337,\n          3.0,\n
150.0\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"petal.length\",\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 51.83521261418364,\n        \"min\":
1.0,\n        \"max\": 150.0,\n        \"num_unique_values\": 8,\n
\"samples\": [\n          3.7580000000000005,\n          4.35,\n
150.0\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"petal.width\",\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 52.636648242617504,\n        \"min\":
0.1,\n        \"max\": 150.0,\n        \"num_unique_values\": 8,\n
\"samples\": [\n          1.1993333333333336,\n          1.3,\n
150.0\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    }\n  ]\n}","type":"dataframe"}
```


```
data.value_counts('variety')
```

```
variety
Setosa        50
Versicolor    50
Virginica     50
Name: count, dtype: int64
```

```
sns.countplot(x='variety',data=data,)
plt.show()
```

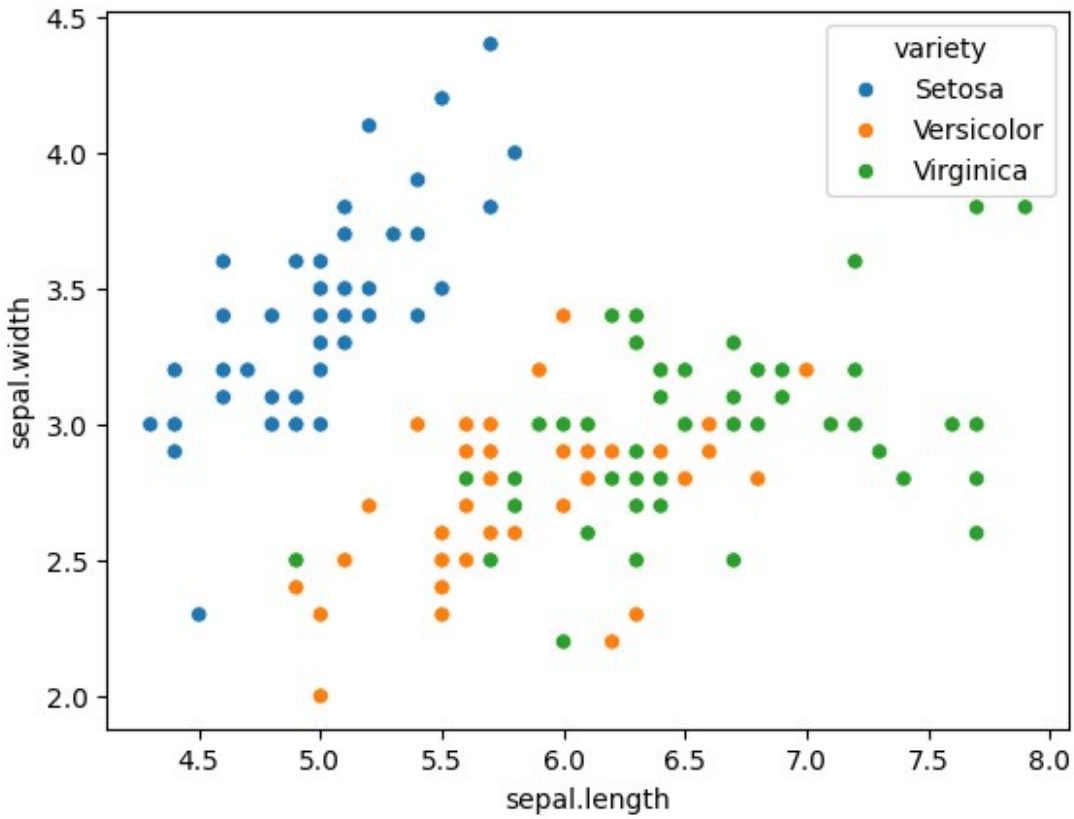



```
dummies=pd.get_dummies(data.variety)
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,
[0,1,2,3]]],axis=1)
FinalDataset.head()
```

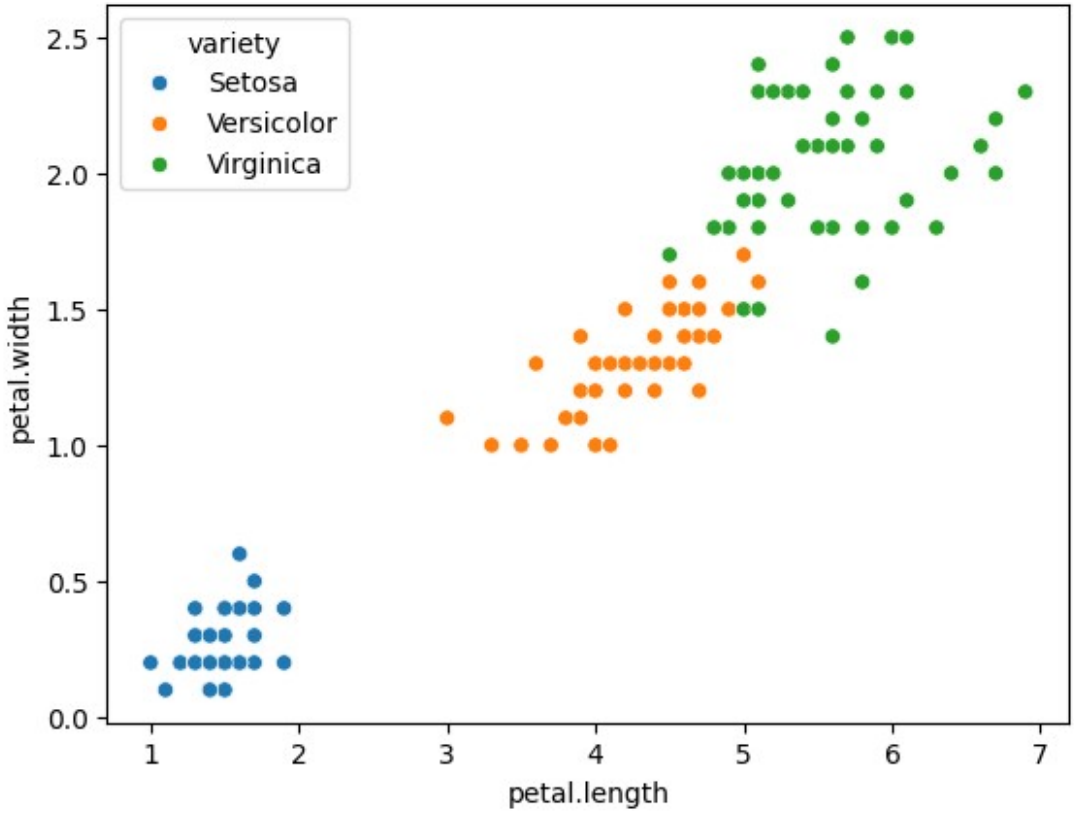

```
{"summary":"{\n  \"name\": \"FinalDataset\",\n  \"rows\": 150,\n
\"fields\": [\n    {\n      \"column\": \"Setosa\",\n
\"properties\": {\n        \"dtype\": \"boolean\",\n
\"num_unique_values\": 2,\n        \"samples\": [\n          false,\n
true\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"Versicolor\",\n      \"properties\": {\n        \"dtype\":
\"boolean\",\n        \"num_unique_values\": 2,\n        \"samples\":
[\n          true,\n          false\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Virginica\",\n
\"properties\": {\n        \"dtype\": \"boolean\",\n
\"num_unique_values\": 2,\n        \"samples\": [\n          true,\n
false\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"sepal.length\",\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 0.8280661279778629,\n        \"min\":
4.3,\n        \"max\": 7.9,\n        \"num_unique_values\": 35,\n
\"samples\": [\n          6.2,\n          4.5\n        ],\n
```

```
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"sepal.width\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
0.435866284936698,\n        \"min\": 2.0,\n        \"max\": 4.4,\n
\"num_unique_values\": 23,\n        \"samples\": [\n          2.3,\n
4.0\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"petal.length\",\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 1.7652982332594667,\n        \"min\":
1.0,\n        \"max\": 6.9,\n        \"num_unique_values\": 43,\n
\"samples\": [\n          6.7,\n          3.8\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"petal.width\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
0.7622376689603465,\n        \"min\": 0.1,\n        \"max\": 2.5,\n
\"num_unique_values\": 22,\n        \"samples\": [\n          0.2,\n
1.2\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    }\n  ]\
n}","type":"dataframe","variable_name":"FinalDataset"}
```

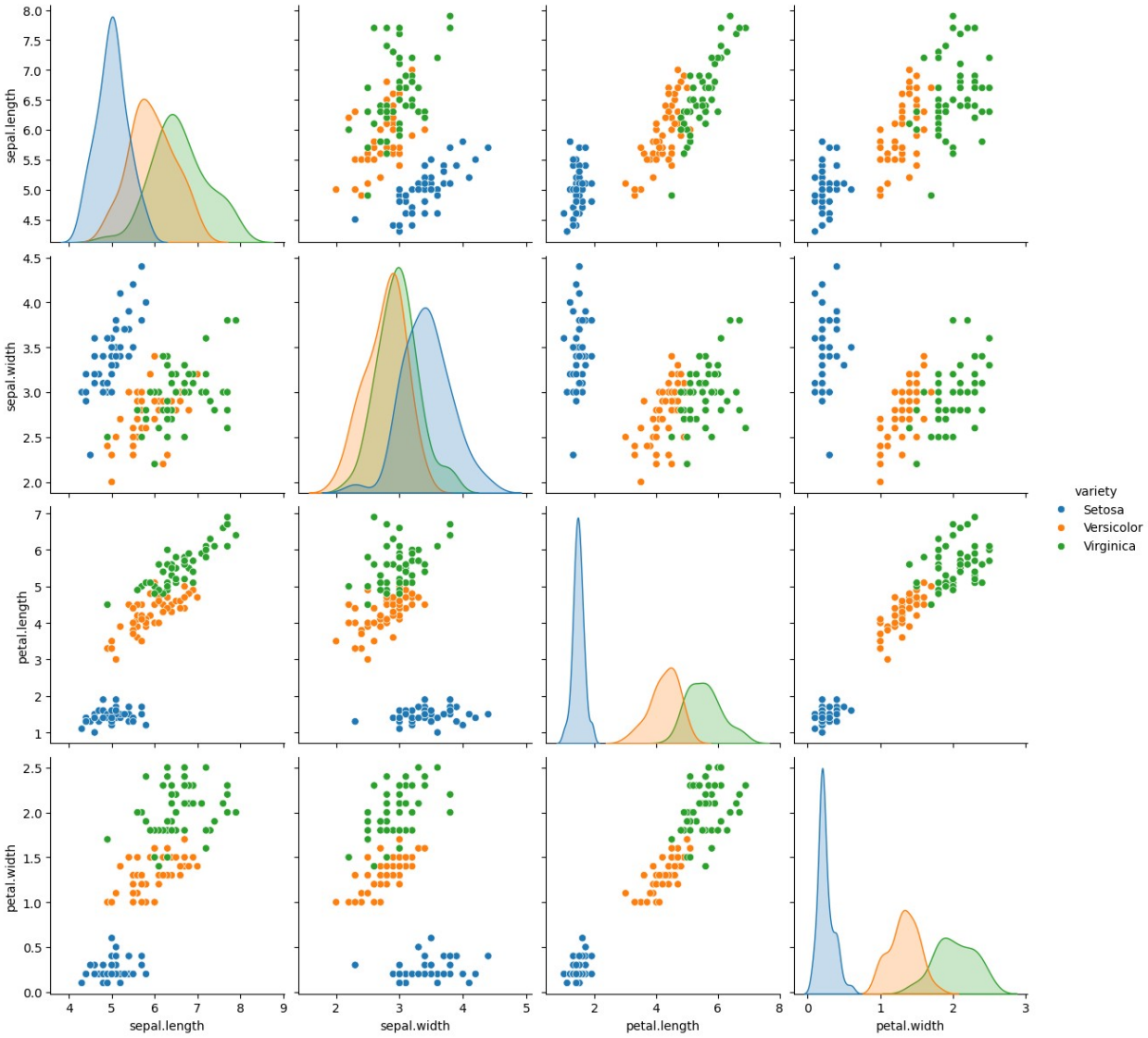

```
sns.scatterplot(x='sepal.length', y='sepal.width', hue='variety',
data=data)
plt.show()
```



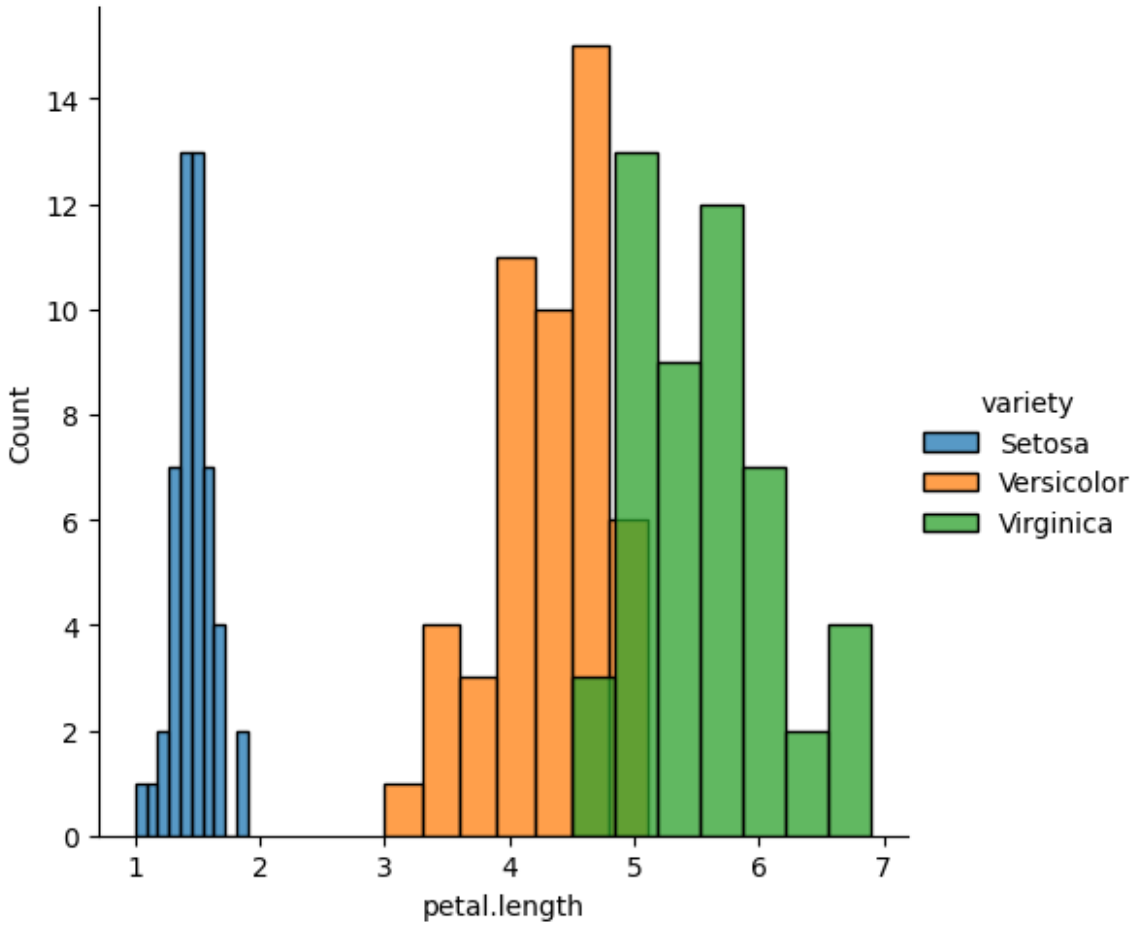
```
sns.scatterplot(x='petal.length', y='petal.width', hue='variety',
data=data)
plt.show()
```



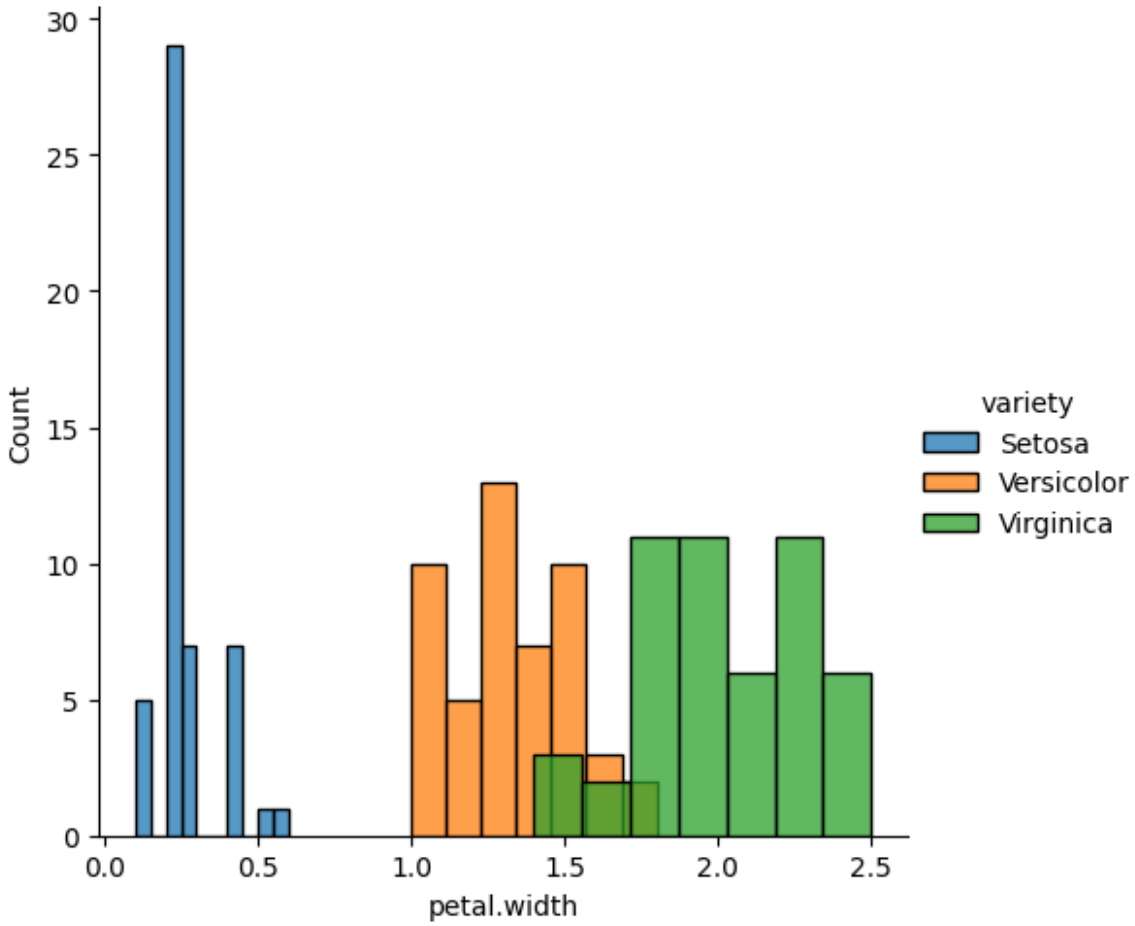

```
sns.pairplot(data,hue='variety',height=3);
```



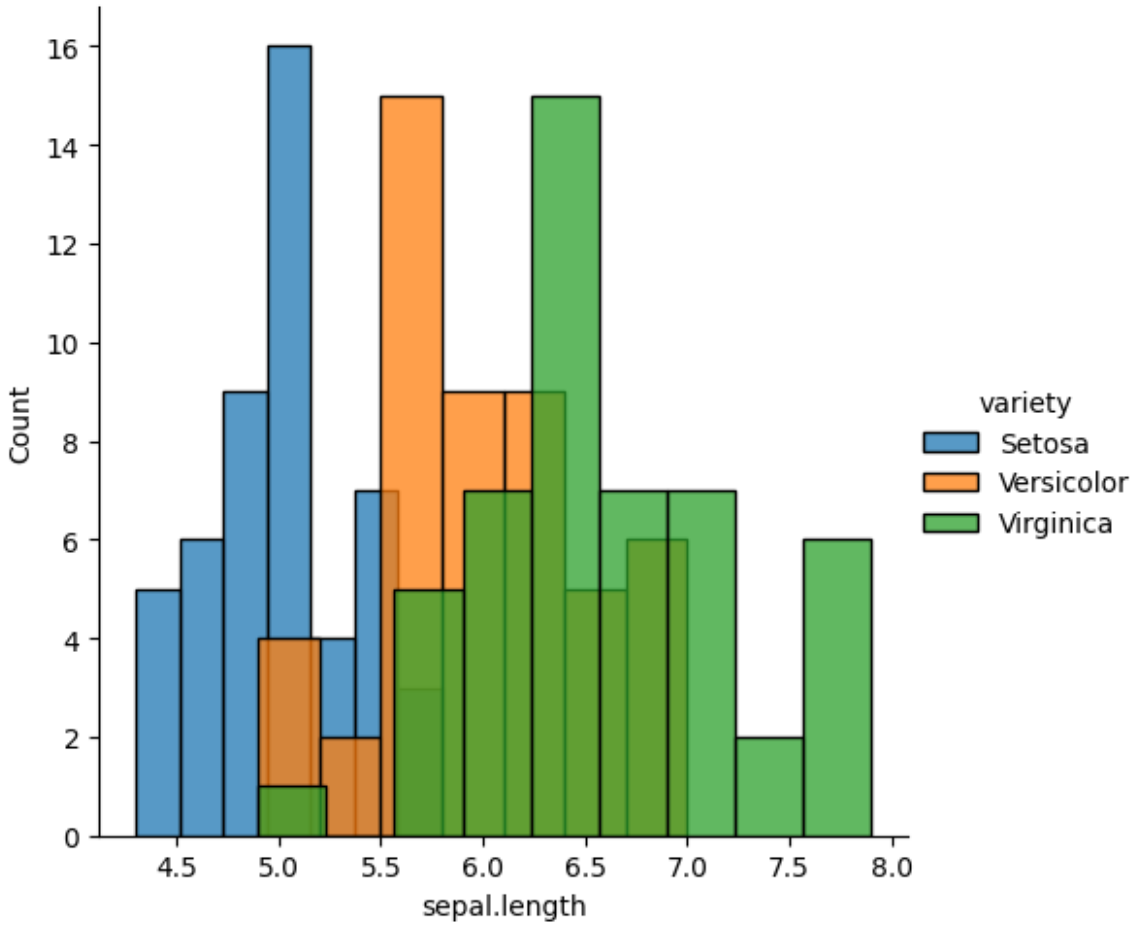

```
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'petal.length').add_legend();
plt.show();
```



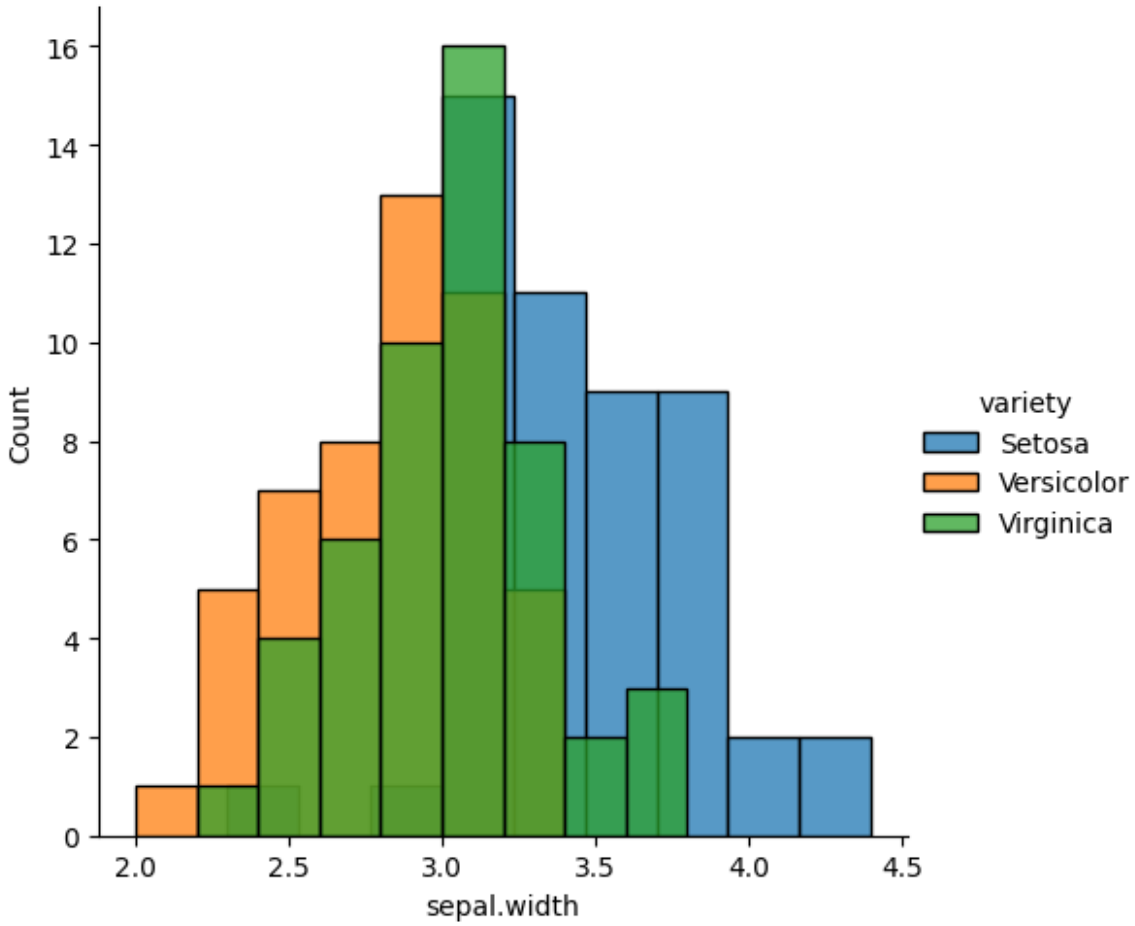

```
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'petal.width').add_legend();
plt.show();
```




```
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'sepal.length').add_legend();
plt.show();
```




```
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'sepal.width').add_legend();
plt.show();
```


Count
sepal.width
variety
Setosa
Versicolor
Virginica
0
2
4
6
8
10
12
14
16
2.0
2.5
3.0
3.5
4.0
4.5